

Discrepancies between TIBU and clinics' paper records: Evidence from a secondary analysis of a 2018–19 audit

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Since its introduction in 2012, Kenya's TIBU database has increasingly served as the source for peer-reviewed tuberculosis research in Kenya, including national analyses of treatment interruption and mortality^{1,2}, HIV–TB co-infection outcomes^{3,4}, pediatric TB epidemiology and forecasting^{5,6}, drug-resistant TB⁷, county-level retrospective cohorts of treatment outcomes and mortality^{8–11}, community health volunteers' contribution to case notification¹², and national estimates of TB under-reporting¹³. Despite the growing reliance on TIBU, its accuracy remains relatively under-studied. A high quality audit of $N = 222$ cases registered in 2012 found that, three years later (in 2015) TIBU was more complete than TB treatment cards and treatment clinic registries. However, when a patient's paper records could be located, they often did not concord with TIBU, e.g., over 30% of outcome data in TIBU did not concord with the outcomes in the paper records¹⁴. In subsequent years, TIBU's recording and reporting tools were revised and a recurring data-quality-assessment (DQA) programme was instituted^{15,16}; the most recent DQA (2022) reports high concordance between TIBU and the facility TB register ($\sim 89\%$ of treatment outcomes, Cohen's $\kappa \approx 0.86$)¹⁶. These assessments are internal program self-audits on small, purposively selected samples, however, and TIBU has not since been the subject of an independent, systematically sampled audit.

This study helps to fill that gap. It presents exploratory, secondary analyses of an audit of $N = 4,570$ TIBU records, which was conducted as part of a trial evaluating a digital adherence tool called Keheala¹⁷. Although a relatively large number of records was audited, we note up-front that the audit had important limitations: it was restricted to a subset of the Keheala trial participants, and focused exclusively on outcome data; no effort was made to systematically evaluate TIBU beyond this population, or to comprehensively digitize other fields in the registry. However, despite these limitations, we believe analyses of the audit data can be informative to researchers who rely on TIBU to study tuberculosis in Kenya.

1 Methods

1.1 Data sources and audit sample

The TIBU extract examined herein covers patients registered between April 2018 and December 2019 ($N = 119,653$ nationwide), the Keheala trial study period. The data were extracted in December 2021, and treatment outcomes were recorded through late 2021. The long outcome window reflects standard TB treatment duration: roughly six months for drug-sensitive disease and up to 18–24

months for multidrug-resistant (MDR) disease.

The Keheala trial was restricted to the following counties: Kakamega, Kiambu, Kisumu, Machakos, Mombasa, Nairobi, Turkana, and Wajir. Additional restrictions included: access to a cell phone, and knowledge of English or Kiswahili¹⁷. Clinic participation in the Keheala trial was voluntary. Within the study period, 16,579 patients were registered in clinics that participated in the trial.

For the audit, Wajir was omitted because it is remote and difficult to reach, and had few participants. For each of the remaining seven counties, the Keheala trial study team digitized paper-registry records for all Keheala trial participants in two sub-counties. These sub-counties were selected via the following cluster-randomization procedure. Within each county, sub-counties were ordered by the number of participants, and a median split was performed, so that one 'block' of sub-counties comprised a relatively small group of the largest sub-counties and the other block a larger group of smaller sub-counties. One sub-county was then drawn from each block to represent the county (two in total: one large, one small), for 14 audited sub-counties in all.¹ All 4,570 trial participants registered in these sub-counties during the study period constitute the population from which paper records were sought (the *selected* sample).

The study team searched clinics' paper registries for these participants in multiple waves, beginning in late 2018 and ending in mid-2020, matching records on SCRN and spot-checking against patient and disease characteristics such as age, sex, and HIV co-infection. Figure 1 summarizes the resulting exclusion cascade. Of the 4,570 selected participants, 3,497 were *located* in the paper registry; the remaining 1,073 could not be found and are not auditable. From the located records we exclude only the 191 patients whose paper record contains no classifiable outcome (blank or the interim "not completed" code), as the audit requires a ground-truth outcome. This leaves an *analyzed* sample of 3,306 patients.

¹We identify the audited sub-counties operationally as the two sub-counties per county containing the most patients located in the paper audit; this reproduces the two-per-county design and excludes a small number of stray, cross-coded patients in neighbouring sub-counties.

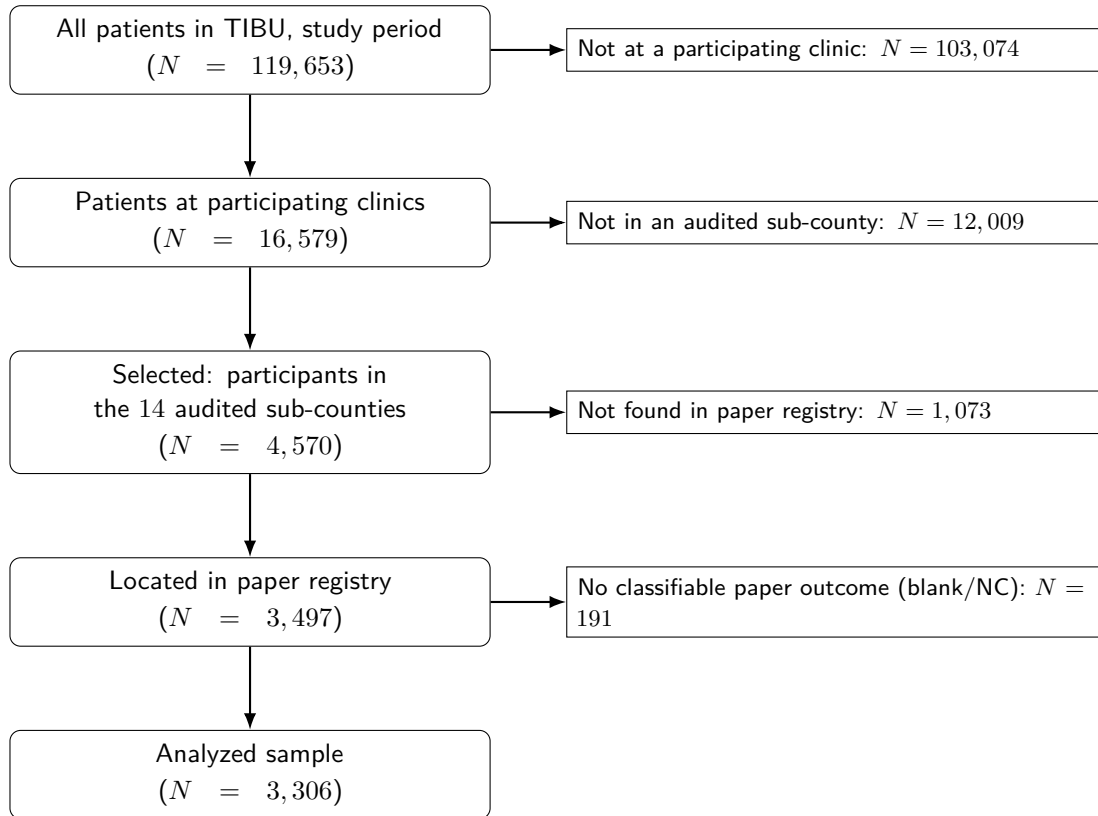


Figure 1: Exclusion cascade from all patients registered in TIBU during the study period to the analyzed sample; the box to the right of each step gives the number excluded at that step.

The individual and disease characteristics reported in Tables 1a and 1b are drawn from the TIBU record and defined as follows: a patient is *bacteriologically confirmed* if the baseline (month-0) sputum smear is positive or GeneXpert reports “MTB detected”; *drug resistant* if a resistance pattern is recorded; *extrapulmonary* if the disease site is coded extrapulmonary; a *retreatment* case if the registration type is relapse or treatment after loss to follow-up; and *HIV positive* if HIV status is recorded positive.

1.2 Defining and Classifying Discrepancies

The central outcome of interest is a discrepancy between a patient’s record in TIBU and in their clinic’s TB register, as digitized by the study team. A record whose TIBU and paper outcomes agree is *concordant*; a record is a *discrepancy* if the two outcomes disagree, or if the TIBU outcome is missing (blank or the interim “not completed” code) while the paper register records one.

TIBU is a paper-to-digital reporting system: clinic staff record treatment data on the paper TB register at the point of care, and TB clerks subsequently transcribe those entries into TIBU. We therefore treat the paper register as ground truth, so that a discrepancy reflects a gap or error in the transcription into TIBU.

For many of our analyses, we classify discrepancies into categories as follows. First, we assign every treatment outcome—in both TIBU and the paper register—to one of three documented states:

treatment *success* (cure or treatment completion), treatment *failure* (death, treatment failure, or loss to follow-up), or *transfer-out*. Transfer-out (including the MT4 code, used when drug resistance is discovered mid-treatment and the patient is referred to a drug-resistant TB program) is a documented outcome rather than an exclusion: a transfer recorded in the paper register is ground truth like any other outcome, and a TIBU value that disagrees with it is a countable discrepancy.

Then, comparing the TIBU-recorded outcome against the paper register yields four categories of discrepancy, each named after the outcome TIBU records when it disagrees with the paper register:

- **False positive** (a Type I error): TIBU records a successful outcome but the paper registry records something else—a failure or a transfer-out. TIBU *overstates* the program’s success.
- **False negative** (a Type II error): TIBU records a failure but the paper registry records something else—a success or a transfer-out. TIBU *understates* the true status.
- **False transfer**: TIBU records the patient as transferred out, but the paper registry records a real success or failure—a patient who was in fact seen through to an outcome at the clinic.
- **Missing in TIBU**: TIBU contains no outcome (blank or the interim “not completed” code) but the paper registry records one. This is distinct from loss-to-follow-up, which is itself a recorded outcome: the clinic did record an outcome on paper, but the TIBU entry was never created or finalized.

1.3 Statistical analysis

Our analyses proceed in three steps. First, the cross-tabulation in Table 2 gives the full distribution of outcome discrepancies across the four categories defined above. Second, we break these discrepancies down by *when* they occurred (a smoothed time series over the study period, §2.2.2), *where* they occurred (clinic-level, by province, urbanicity, and size, §2.2.3), and among *whom* (patient-level, by sex, age, disease type, and HIV status, §2.2.4); a multivariate logistic-regression counterpart of the where/who breakdowns, reporting average marginal effects with clinic-clustered standard errors, appears in Appendix D. Third, we use the records’ date fields as a sensitivity check on these discrepancy counts (§2.3).

The time-series smoothing (§2.2.2) fits each record’s binary discrepancy indicator against its outcome date by locally-weighted regression (LOWESS, span 0.35), with pointwise 95% confidence bands from a percentile bootstrap; the full estimator is described in Appendix A.

A final set of analyses (§2.3) uses the date fields to probe the assumption that the digitized paper record is ground truth, since the audit’s own transcription can err, e.g., through typos, misattribution of a record, or, most relevantly for these analyses, transcription of an out-of-date registry entry. We use three dates per record—the TIBU registration date, the TIBU outcome date, and the paper outcome date—treating implausible values (year before 2010 or after 2025) as missing, and we derive an *outcome date* for the time series (the TIBU outcome date, or the paper outcome date when TIBU is blank, so that missing-in-TIBU records remain plottable) and a *record age* (days from registration to a fixed extract snapshot of 1 January 2022) used as a covariate. We first compare the TIBU and paper outcome dates directly, by discrepancy category (Table 5 and Figure 3). We then use these dates to construct a *date-gated* reclassification of discrepancies: a TIBU–paper disagreement is

counted as a discrepancy only when the paper outcome date is strictly later than the TIBU outcome date, and is otherwise treated as concordant, on the reading that TIBU carries a post-audit update to the register. Because the audit transcribed dates but did not prioritize their accuracy, these date-field analyses should not be interpreted as conclusive.

1.4 Preregistration

No analyses were pre-registered for this study. The Keheala trial's protocol specified the sampling procedure described above, and the trial's own analyses of the audit data focused on overall false positive and false negative error rates, as well as comparisons of these rates across the trial's treatment groups.

2 Results

2.1 Study Population

Tables 1a and 1b describe the study population and its TIBU-recorded outcomes across the five nested cohorts of the exclusion cascade (Figure 1): all patients registered in TIBU during the study period; all patients at participating clinics; the *selected* participants in the audited sub-counties; those *located* in the paper registry; and the *analyzed* sample. The distributions are stable across the cohorts, indicating that the audited sample is broadly representative of the wider registry population.

Table 1a: Individual and disease characteristics across the five nested cohorts of the exclusion cascade. All values are from the TIBU digital record.

	All TIBU (N=119,653)		Participating clinics (N=16,579)		Selected (N=4,570)		Located (N=3,497)		Analyzed (N=3,306)	
	N	(%)	N	(%)	N	(%)	N	(%)	N	(%)
<i>Individual characteristics</i>										
Male	75,597	(63.2%)	10,406	(64.5%)	2,853	(64.1%)	2,184	(64.2%)	2,115	(64.3%)
Mean age (years)	36.1		34.3		33.3		32.8		32.6	
<i>Disease characteristics</i>										
Bacteriologically confirmed	62,149	(51.9%)	12,080	(74.9%)	3,239	(72.8%)	2,410	(70.8%)	2,326	(70.7%)
Drug resistant	1,343	(1.1%)	17	(0.1%)	4	(0.1%)	3	(0.1%)	2	(0.1%)
Extrapulmonary TB	17,558	(14.7%)	1,992	(12.4%)	476	(10.7%)	383	(11.3%)	367	(11.2%)
HIV coinfection	32,077	(26.8%)	4,212	(26.4%)	1,180	(27.0%)	760	(22.7%)	729	(22.6%)
Retreatment	8,351	(7.0%)	1,337	(8.5%)	434	(10.0%)	307	(9.2%)	298	(9.2%)

Table 1b: TIBU-recorded treatment outcomes across the five nested cohorts of the exclusion cascade.

	All TIBU (N=119,653)		Participating clinics (N=16,579)		Selected (N=4,570)		Located (N=3,497)		Analyzed (N=3,306)	
	N	(%)	N	(%)	N	(%)	N	(%)	N	(%)
<i>Outcomes</i>										
Cured	25,399	(21.2%)	6,724	(40.6%)	1,578	(34.5%)	1,144	(32.7%)	1,103	(33.4%)
Treatment completed	31,719	(26.5%)	6,610	(39.9%)	1,963	(43.0%)	1,520	(43.5%)	1,490	(45.1%)
Died	4,911	(4.1%)	764	(4.6%)	231	(5.1%)	184	(5.3%)	180	(5.4%)
Treatment failed	382	(0.3%)	123	(0.7%)	46	(1.0%)	34	(1.0%)	34	(1.0%)
Lost to follow-up	3,594	(3.0%)	678	(4.1%)	192	(4.2%)	131	(3.7%)	128	(3.9%)
Not completed	47,184	(39.4%)	562	(3.4%)	130	(2.8%)	113	(3.2%)	89	(2.7%)
Transfer out	6,382	(5.3%)	620	(3.7%)	286	(6.3%)	264	(7.5%)	257	(7.8%)
Missing	82	(0.1%)	498	(3.0%)	144	(3.2%)	107	(3.1%)	25	(0.8%)

2.2 Outcome Discrepancies

2.2.1 Crosstab of TIBU vs. Paper Registry Outcomes

Table 2 cross-tabulates treatment outcomes recorded in TIBU against those recorded in the paper registry, for all located patients ($N = 3,497$). Concordant records lie on the diagonal; the off-diagonal cells correspond to discrepancies. Across the 3,306 records with a classifiable paper outcome, 410 (12.4%) are discrepancies. Those discrepancies that fall into our four categories are shaded: false positive (red), false negative (green), missing-in-TIBU (blue), and false transfer (orange).

Table 2: Cross-tabulation of treatment outcomes in TIBU vs. paper registry.

Outcome in TIBU	Treatment Outcome in Paper Registry										Total
	Blank	C	D	F	LTFU	NC	TC	TO			
	<i>N</i>	<i>N</i>	<i>N</i>	<i>N</i>	<i>N</i>	<i>N</i>	<i>N</i>	<i>N</i>	<i>N</i>	<i>N</i>	
Blank	82 (76.6%)	14 (13.1%)	1 (0.9%)	0 (0.0%)	2 (1.9%)	0 (0.0%)	6 (5.6%)	2 (1.9%)			107
C	41 (3.6%)	995 (87.0%)	1 (0.1%)	3 (0.3%)	7 (0.6%)	0 (0.0%)	55 (4.8%)	42 (3.7%)			1,144
D	4 (2.2%)	2 (1.1%)	166 (90.2%)	0 (0.0%)	1 (0.5%)	0 (0.0%)	8 (4.3%)	3 (1.6%)			184
F	0 (0.0%)	1 (2.9%)	1 (2.9%)	25 (73.5%)	5 (14.7%)	0 (0.0%)	2 (5.9%)	0 (0.0%)			34
LTFU	3 (2.3%)	2 (1.5%)	1 (0.8%)	0 (0.0%)	113 (86.3%)	0 (0.0%)	6 (4.6%)	6 (4.6%)			131
NC	23 (20.4%)	34 (30.1%)	6 (5.3%)	0 (0.0%)	4 (3.5%)	1 (0.9%)	32 (28.3%)	13 (11.5%)			113
TC	30 (2.0%)	76 (5.0%)	7 (0.5%)	0 (0.0%)	12 (0.8%)	0 (0.0%)	1,333 (87.7%)	62 (4.1%)			1,520
TO	7 (2.7%)	45 (17.0%)	5 (1.9%)	5 (1.9%)	14 (5.3%)	0 (0.0%)	63 (23.9%)	125 (47.3%)			264
Total	190	1,169	188	33	158	1	1,505	253			3,497

Note: Percentages are row shares. Transfer-out (TO) is a documented outcome; a disagreeing TIBU value is a discrepancy named after what TIBU recorded.

Red = False positive (TIBU: success; paper: failure or transfer).

Green = False negative (TIBU: failure; paper: success or transfer).

Orange = False transfer (TIBU: transfer; paper: a real success/failure).

Blue = Missing in TIBU (TIBU: no outcome; paper: outcome present).

2.2.2 When did discrepancies occur?

Figure 2 plots each discrepancy rate over the study period (LOWESS smooth with percentile-bootstrap 95% bands; Appendix A).

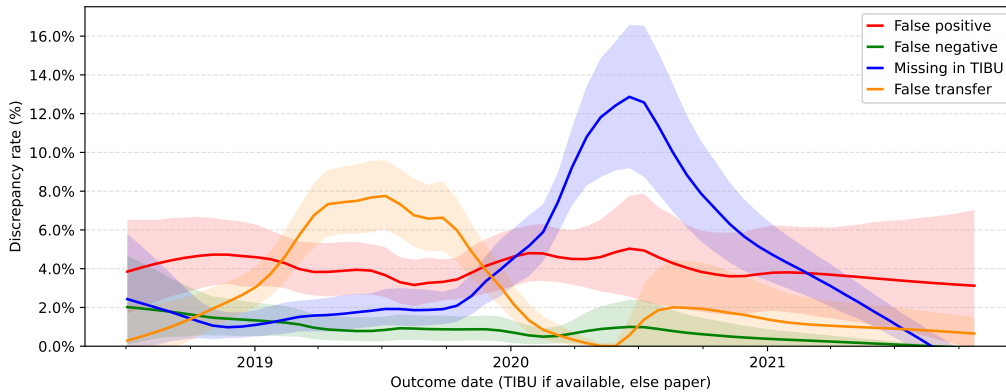


Figure 2: DQA discrepancy rates over time by outcome date (TIBU when available, paper otherwise). Lowess smooth with percentile bootstrap 95% CI bands.

The missing-in-TIBU rate peaks at roughly 15% for records with outcome dates in 2020 Q1–Q2; 59 of the 114 missing-in-TIBU records fall in these two quarters. Of all 114 missing-in-TIBU records, 89 carry the interim “not completed” code and 25 are entirely blank. False-positive and false-negative

rates are low and roughly flat across the study period. False transfers are concentrated among records registered in 2019 (0.4% of 2018 registrations versus 6.2% of 2019) and peak at 2019 outcome dates.

Broadly speaking, these patterns are also present when records are placed by their paper outcome date rather than TIBU's (Appendix B).

2.2.3 Where did discrepancies occur?

Table 3 reports the four discrepancy rates stratified by province, urbanicity, and clinic size. Missing-in-TIBU is elevated at small clinics, in rural areas, and in two provinces (Eastern South and Rift Valley South); false transfers are elevated in Nairobi and at larger clinics.

Most discrepancies are concentrated in a handful of clinics (Table 3b). Missing-in-TIBU is concentrated in clinics 0 and 115 (rates of 14.1% and 15.6%, roughly four times the study-wide 3.4%), which contribute 60 of the 114 missing-in-TIBU records. False transfers are concentrated in clinics 162 and 50, both in Nairobi (rates above 20%, five times the study-wide average). False positives are concentrated in clinic 482 alone (63 of the 134).

In multivariate models of missing-in-TIBU and false transfer (Appendix D), the associations with the share of male patients and with clinic size become non-significant once each outcome's two dominant clinics are excluded; only the association with record age partially remains.

Table 3: DQA discrepancy rates by province, urbanicity, and clinic size.

	<i>N</i>	(%)	False positive <i>N</i>	(%)	False negative <i>N</i>	(%)	Missing in TIBU <i>N</i>	(%)	False transfer <i>N</i>	(%)
<i>By province</i>										
Central	27	(0.8%)	0	(0.0%)	0	(0.0%)	13	(48.1%)	0	(0.0%)
Coast	708	(21.4%)	18	(2.5%)	4	(0.6%)	13	(1.8%)	6	(0.8%)
Eastern North	2	(0.1%)	0	(0.0%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Eastern South	264	(8.0%)	8	(3.0%)	2	(0.8%)	24	(9.1%)	12	(4.5%)
Nairobi	535	(16.2%)	18	(3.4%)	7	(1.3%)	4	(0.7%)	89	(16.6%)
North Eastern	821	(24.8%)	66	(8.0%)	6	(0.7%)	2	(0.2%)	19	(2.3%)
Nyanza North	392	(11.9%)	6	(1.5%)	7	(1.8%)	3	(0.8%)	2	(0.5%)
Nyanza South	40	(1.2%)	1	(2.5%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Rift Valley North	44	(1.3%)	0	(0.0%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Rift Valley South	425	(12.9%)	15	(3.5%)	3	(0.7%)	37	(8.7%)	4	(0.9%)
Western	31	(0.9%)	2	(6.5%)	1	(3.2%)	1	(3.2%)	0	(0.0%)
<i>By urban/rural</i>										
Urban	1,118	(33.8%)	30	(2.7%)	10	(0.9%)	21	(1.9%)	95	(8.5%)
Rural	2,167	(65.5%)	104	(4.8%)	20	(0.9%)	76	(3.5%)	37	(1.7%)
<i>By clinic size</i>										
Large clinic	1,273	(38.5%)	39	(3.1%)	10	(0.8%)	24	(1.9%)	90	(7.1%)
Small clinic	2,033	(61.5%)	95	(4.7%)	20	(1.0%)	90	(4.4%)	42	(2.1%)
All	3,306	(100.0%)	134	(4.1%)	30	(0.9%)	114	(3.4%)	132	(4.0%)

Note: Urban/rural classification is based on clinic records. Large clinics are those with at least 7 patients registered in TIBU (median); small clinics are those with fewer.

Table 3b: Clinics in which discrepancies are concentrated. Discrepancies in the five clinics accounting for most discrepancies (*N* and %, the share of all records in that clinic containing a discrepancy). Discrepancies in all remaining clinics and the study-wide average are shown for reference.

Clinic	Province	<i>N</i>	False positive <i>N</i> (%)	False negative <i>N</i> (%)	Missing in TIBU <i>N</i> (%)	False transfer <i>N</i> (%)
0	Rift Valley S.	263	9 (3.4%)	0 (0.0%)	37 (14.1%)	3 (1.1%)
115	Eastern S.	147	2 (1.4%)	0 (0.0%)	23 (15.6%)	11 (7.5%)
162	Nairobi	242	9 (3.7%)	3 (1.2%)	3 (1.2%)	49 (20.2%)
50	Nairobi	170	3 (1.8%)	1 (0.6%)	1 (0.6%)	38 (22.4%)
482	North Eastern	806	63 (7.8%)	6 (0.7%)	2 (0.2%)	18 (2.2%)
<i>All other clinics</i>		1,678	48 (2.9%)	20 (1.2%)	48 (2.9%)	13 (0.8%)
<i>Study-wide</i>		3,306	134 (4.1%)	30 (0.9%)	114 (3.4%)	132 (4.0%)

2.2.4 Who experienced discrepancies?

Table 4 reports the discrepancy rates by individual and disease characteristics. Missing-in-TIBU is elevated among male patients (3.8%), the 15–34 age band (2.8%), and HIV-coinfected patients (2.9%). In the multivariate model, no patient characteristic independently predicts missingness once the two clinics with the highest missing-in-TIBU rates are excluded, and the excess of false transfers among male patients does not persist once the two Nairobi clinics with the highest false-transfer rates are excluded (Appendix D). False-positive and false-negative rates show no notable associations with individual or disease characteristics.

Table 4: DQA discrepancy rates by individual and disease characteristics.

	<i>N</i>	(%)	False positive <i>N</i>	(%)	False negative <i>N</i>	(%)	Missing in TIBU <i>N</i>	(%)	False transfer <i>N</i>	(%)
<i>Individual characteristics</i>										
Male	2,115	(64.0%)	76	(3.6%)	18	(0.9%)	81	(3.8%)	103	(4.9%)
<i>Age group</i>										
<15	458	(13.9%)	13	(2.8%)	3	(0.7%)	7	(1.5%)	13	(2.8%)
15–34	1,528	(46.2%)	69	(4.5%)	15	(1.0%)	43	(2.8%)	67	(4.4%)
35–54	965	(29.2%)	32	(3.3%)	10	(1.0%)	32	(3.3%)	44	(4.6%)
55+	331	(10.0%)	20	(6.0%)	2	(0.6%)	9	(2.7%)	8	(2.4%)
<i>Disease characteristics</i>										
Bacteriologically confirmed	2,326	(70.4%)	104	(4.5%)	20	(0.9%)	71	(3.1%)	105	(4.5%)
Extrapulmonary	367	(11.1%)	11	(3.0%)	3	(0.8%)	9	(2.5%)	16	(4.4%)
Retreatment	298	(9.0%)	16	(5.4%)	2	(0.7%)	5	(1.7%)	6	(2.0%)
HIV positive	729	(22.1%)	28	(3.8%)	14	(1.9%)	21	(2.9%)	21	(2.9%)
All	3,306	(100.0%)	134	(4.1%)	30	(0.9%)	114	(3.4%)	132	(4.0%)

Note: Each row shows the count and rate of records matching the row label, and the count and rate of each discrepancy type within those records. Rows are not mutually exclusive (e.g. a male HIV-positive patient appears in both rows).

2.3 Incorporating Date Fields to Assess Outcome Discrepancies

We compare the TIBU and paper outcome dates for the DQA-matched records with parseable dates on both sides; the audit transcribed dates for 3,186 of the 3,645 records (87.4%). Table 5 and

Figure 3 report the comparison.

Table 5: TIBU–paper outcome-date agreement, by outcome-discrepancy category.

Category	<i>N</i>	TIBU – paper (days)					TIBU date is	
		≤ 7 days	8–30 days	31–90 days	91–180 days	>180 days	Earlier	Later
Concordant	2,766	2140 (77.4%)	203 (7.3%)	136 (4.9%)	51 (1.8%)	236 (8.5%)	371 (13.4%)	450 (16.3%)
False positive	57	9 (15.8%)	4 (7.0%)	6 (10.5%)	24 (42.1%)	14 (24.6%)	6 (10.5%)	43 (75.4%)
False negative	21	2 (9.5%)	3 (14.3%)	5 (23.8%)	7 (33.3%)	4 (19.0%)	14 (66.7%)	6 (28.6%)
False transfer	130	8 (6.2%)	7 (5.4%)	6 (4.6%)	50 (38.5%)	59 (45.4%)	117 (90.0%)	6 (4.6%)

Note: Denominator is records where both the TIBU and paper outcome dates are non-blank and parseable. Missing-in-TIBU records cannot appear (TIBU date is blank by construction); their temporal pattern is in Figure 2. “TIBU earlier” and “TIBU later” together sum to < 100% because some records agree exactly.

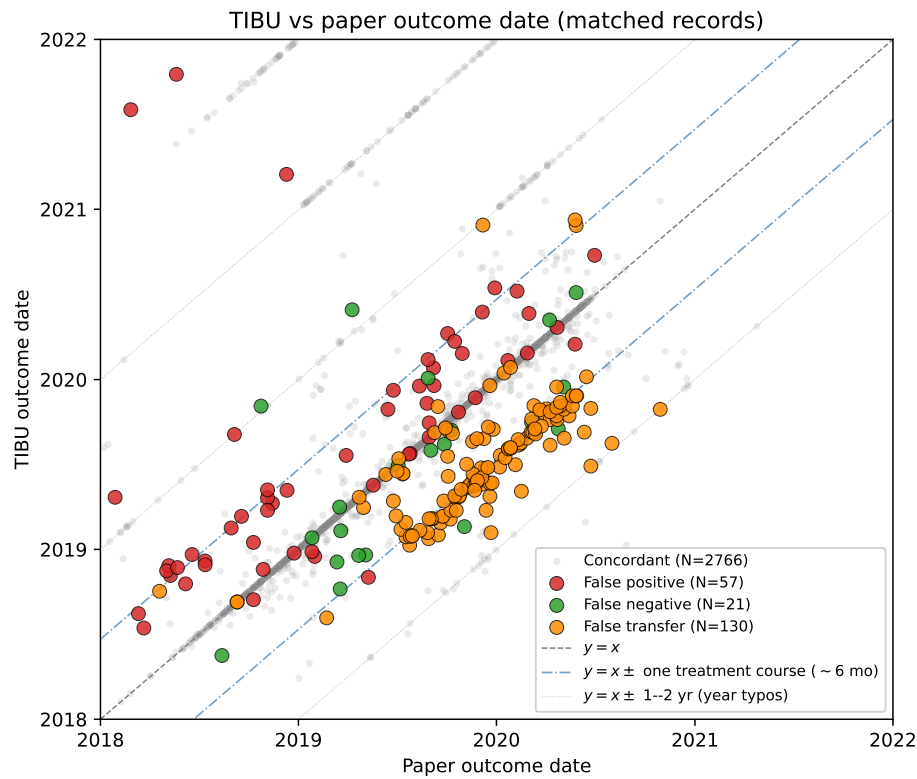


Figure 3: TIBU outcome date vs. paper outcome date for DQA-matched records with parseable dates on both sides (axes clipped to the 2018–2022 study window). Points on the $y = x$ dashed line agree exactly. Two distinct families of parallel structure appear. The coloured discrepancy bands lie one treatment course (~ 6 months) off the $y = x$ line (dash-dot reference lines): transfer-driven false positives (red) a course *above* it and false transfers (orange) a course *below*, mirror-image signatures of the transfer “outcome date” being pinned to registration. The fainter grey streaks lie one to two years off (dotted reference lines), where TIBU and the paper audit disagree on the record’s year without changing the recorded outcome. False-negative points (green) sit modestly below the line. Missing-in-TIBU records cannot appear (TIBU has no outcome date); see Figure 2 for their temporal pattern.

Among concordant records the two dates agree to within a week $\sim 78\%$ of the time. The discrepancy

classes differ (Table 5): for false positives, TIBU is *later* than the paper date about three-quarters of the time; for false negatives, TIBU is *earlier* about 70% of the time; and for false transfers, TIBU is earlier in almost all cases ($\sim 95\%$), typically by about six months.

Among the 167 transfer-out disagreements—records where one register recorded a transfer-out and the other a real outcome—the transfer-out “outcome date” falls a median of 0 days after registration (interquartile range ± 18 days), whereas the real outcome on the other side falls a median of 172 days after—one treatment course.

Under the date-gated classification (a disagreement counts as a discrepancy only when the paper outcome date is strictly later than the TIBU outcome date), the false-positive rate falls from 4.05% to 2.75%, the false-negative and false-transfer rates fall only slightly (from 0.91% to 0.73% and from 3.99% to 3.81%), and missing-in-TIBU is unchanged (3.45%; Table 6). Breakdowns of Tables 3 and 4 and Figure 2 under this definition are in Appendix C.

Table 6: Headline vs. date-gated DQA discrepancy counts and rates.

Discrepancy type	Headline		Date-gated	
	<i>N</i>	Rate	<i>N</i>	Rate
False positive	134	4.05%	91	2.75%
False negative	30	0.91%	24	0.73%
False transfer	132	3.99%	126	3.81%
Missing in TIBU	114	3.45%	114	3.45%

Note: Denominator is the $N = 3,306$ classifiable matched records. Under the date-gated definition, a TIBU–paper outcome disagreement is counted as an error only when the paper outcome date is strictly later than the TIBU outcome date; otherwise the record is treated as concordant on the assumption that TIBU reflects a post-audit paper update. The rule cannot be applied to records with an unparseable date on either side ($n = 77$ false-positive and $n = 9$ false-negative records), and such records preserve their headline classification. Missing-in-TIBU is unaffected since TIBU is blank by construction.

3 Robustness to digitization errors

3.1 Case Study

A researcher who employs a TIBU treatment outcome to develop their study’s outcomes can use the discrepancy rates measured here to gauge the impact of digitization errors on their findings. We now demonstrate such an analysis on the Keheala trial data.

The audit provides a full *confusion matrix* between the TIBU-recorded outcome and the paper-register ground truth (Figure 4): most records are concordant—successes and failures agree 95% and 91% of the time—but a TIBU transfer-out is a real success or failure 51% of the time, and a record missing from TIBU is a real outcome in 87% of cases.

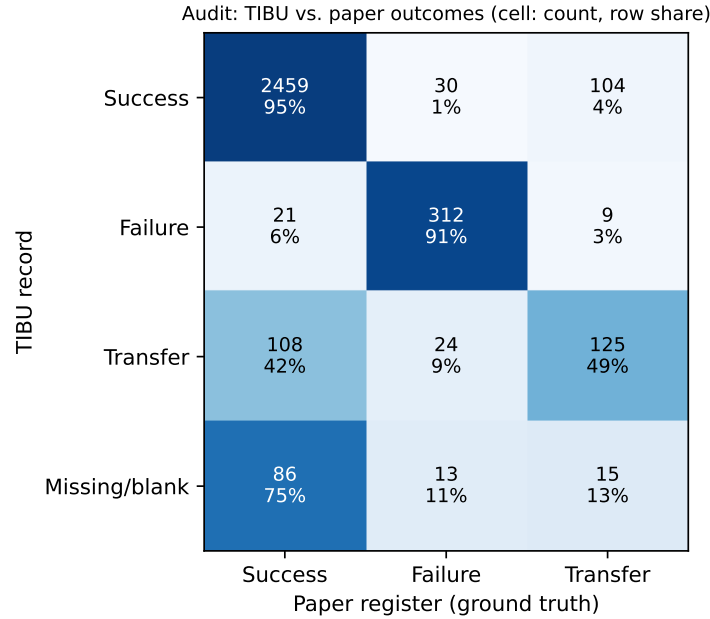


Figure 4: Audit confusion matrix: the TIBU-recorded outcome (rows) against the paper-register ground truth (columns), collapsed to success (C/TC), failure (D/F/LTFU), transfer-out, and—for TIBU only—missing/blank. Each cell gives the count and the row share.

To correct a reported outcome distribution we reclassify a study’s recorded counts through this matrix. Writing n_S, n_F, n_T for the numbers of recorded successes, failures, and transfers, the estimated true counts are $\hat{n}_c = \sum_r n_r P(\text{paper} = c \mid \text{TIBU} = r)$ for $c \in \{S, F, T\}$, and the corrected unsuccessful-outcome share is $\hat{n}_F / (\hat{n}_S + \hat{n}_F)$. This single step absorbs false positives, false negatives, and false transfers together. Where a study also reports records missing from TIBU, the missing row of the matrix is applied to them; because such records are usually absent from the analyzed data, this *false-missing* correction is possible for a trial only if it also includes an audit. Note that this correction assumes *non-differential* misclassification—error rates that do not differ across the groups being compared. This is consistent with the findings of the Keheala trial¹⁷, and reasonable for overall unsuccessful rates or trials that, like the Keheala trial, randomize participants to treatments within clinic.

Among the 14,899 Keheala study-clinic TIBU records with a classifiable success or failure outcome, the reported unsuccessful-outcome share is 10.50%. Reclassifying its successes, failures, and 620 transfer-outs through the matrix raises this to 11.20% at the audit rates (11.92% at twice the rates); applying the missing row to the 1,060 TIBU-missing records raises it marginally further, to 11.31%.

3.2 The impact of digitization errors on prior research

Table 7 applies the same correction to the reported unsuccessful-outcome share of four previous TIBU-based studies, identified in a screen of the works cited in the introduction. For Keheala¹⁷ and Kosgei et al.³ we additionally correct the study’s focal across-group difference—the Keheala-versus-control arm difference and the female-versus-male difference, respectively. In every case the overall rate moves by less than a percentage point, and the focal difference *attenuates* slightly toward zero

(the expected effect of non-differential misclassification) while remaining statistically significant even at twice the audit's error rates.

Table 7: The impact of digitization error correction on the share of unsuccessful outcomes across previous studies that rely on TIBU. Unsuccessful outcomes are D/F/LTFU and successful are C/TC. The overall rate corrects false positives, false negatives, and false transfers by reclassifying each study's outcome counts through the audit confusion matrix (Figure 4); "2×" doubles the off-diagonal (error) rates as a sensitivity bound. The focal across-group difference (percentage points, bootstrap 95% CIs) is the Keheala-versus-control arm difference for Keheala and the female-versus-male difference for Kosgei. [†]Katana et al.'s poor-outcome components sum to 2,398 against a stated 2,408; the gap is immaterial to the correction. [‡]Kosgei et al. analyse a cured-versus-died cohort with no transfer-outs, so only the false-positive/false-negative correction applies and "unsuccessful" denotes death.

Panel A. Overall unsuccessful-outcome rate (%)				
	<i>N</i>	Reported	Corrected	2×
Keheala ¹⁷	14,899	10.50 (9.99, 10.99)	11.20 (10.73, 11.64)	11.92 (11.50, 12.32)
Ngari et al. 2025 ¹¹	36,863	12.67 (12.33, 13.00)	13.13 (12.82, 13.44)	13.61 (13.33, 13.90)
Ngari et al. 2023 ⁴	26,503	13.24 (12.82, 13.65)	13.67 (13.29, 14.05)	14.13 (13.78, 14.47)
Katana et al. 2022 ^{10†}	14,251	13.70 (13.15, 14.28)	14.10 (13.59, 14.63)	14.52 (14.05, 14.99)
Kosgei et al. 2020 ^{3‡}	9,026	9.66 (9.06, 10.28)	10.14 (9.59, 10.71)	10.62 (10.11, 11.15)

Panel B. Focal across-group difference (pp)			
	Reported	Corrected	2×
Keheala vs. control arm ¹⁷	−2.67 (−4.39, −1.06)	−2.43 (−3.94, −0.93)	−2.18 (−3.47, −0.77)
Women vs. men ^{3‡}	−1.78 (−2.95, −0.59)	−1.65 (−2.75, −0.49)	−1.51 (−2.55, −0.47)

4 Discussion

We find that TIBU concords with the clinics' paper registers on 87.6% of treatment outcomes. Discrepancies are concentrated at a small number of clinics and, for records missing from TIBU, in the early-2020 onset of the COVID-19 pandemic. Outright flips between successful (C, TC) and unsuccessful outcomes (LTFU, F, D) are rare: conditionally, a TIBU-recorded success is a failure on-paper just 1.2% of the time and a TIBU-recorded failure a success on-paper 6.3%. Our exploratory analyses using date fields suggest that some of these apparent flips are likely audit artifacts: half of the success-to-failure disagreements (15 of 30) carry an earlier paper date than TIBU, consistent with TIBU holding the later, correct outcome. This lowers the success-to-failure flip rate to ~0.6%.

Particularly for studies that span multiple treatment sites and focus on treatment success, these findings raise confidence in TIBU as a reliable source of data. Our overall concordance rate is much higher than that found by the 2015 TIBU audit¹⁴ and similar to the rates reported by Kenya's NTLD-P¹⁶. The findings of several past studies^{3,4,10,11} are robust to correction for digitization errors.

Our finding that discrepancies are concentrated both temporally and at the clinic level can, we hope, be helpful in further improving TIBU's concordance with clinics' paper registries, as it suggests it is not day-to-day procedures, but unexpected, temporary clinic-level contingencies which most negatively impact TIBU concordance. These findings also mean that digitization correction of between-site

comparisons benefits from site-specific discrepancy rates.

Finally, our experience highlights the benefits of high-fidelity digitization of fields other than outcomes, such as dates, during an audit. These fields are valuable of their own right, and can help when, e.g., determining whether the source of a discrepancy is an error in TIBU, or in the audit itself.

To existing audits, this study adds: independence from NTLD-P, a large sample encompassing many patients across a substantial number of treatment sites with ample temporal variation within clinics, and breakdowns of discrepancies by type. The study's primary limitations are those noted at the outset: our sample was selected from the Keheala trial's¹⁷ participating clinics, which limited its representativeness of all of Kenya's treatment facilities; and it focused on the outcome field alone, which both narrowed the analyses we could undertake and, because other fields—including dates—were not transcribed with the same care, left our date-based analyses exploratory. TIBU would still benefit from a comprehensive, systematically sampled audit.

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Appendix

A Smoothing and confidence bands

The time series of §2.2.2 is estimated nonparametrically. For a given discrepancy type, each analyzed record contributes a binary indicator—1 if the record is that type of discrepancy and 0 otherwise—placed on the horizontal axis by its outcome date (the TIBU outcome date where available, or the paper-registry outcome date for missing-in-TIBU records, which have no TIBU date). We smooth this indicator against outcome date by locally-weighted scatterplot smoothing (LOWESS): at each point a tricube-weighted local linear regression is fit to the nearest 35% of records (smoothing span 0.35, no robustifying iterations), so the fitted value estimates the probability that a record with that outcome date is a discrepancy of the given type—the local discrepancy rate. Pointwise 95% confidence bands are obtained by a percentile bootstrap: the records are resampled with replacement 500 times, the LOWESS curve is refit on each resample and evaluated on a common grid of 60 points spanning the 1st–99th percentiles of the outcome-date range, and the 2.5th and 97.5th percentiles of the bootstrap curves are taken at each grid point; a curve is drawn only where at least 30 records are available. The same estimator produces the paper-date and date-gated sensitivity versions of the figure (Appendices B and C), differing only in the date used for the axis or the discrepancy definition.

B Sensitivity check: paper-registry outcome dates

Figure 5 replays the over-time analysis of §2.2 using the paper-registry outcome date throughout, rather than TIBU's outcome date with paper fallback for missing-in-TIBU records. Records with no parseable paper date are excluded ($\sim 13\%$ of the classifiable sample). The COVID-era peak in missing-in-TIBU is unchanged in shape and magnitude, indicating that it is driven by when outcomes actually occurred (as captured on paper) and not by when TIBU happened to record them.

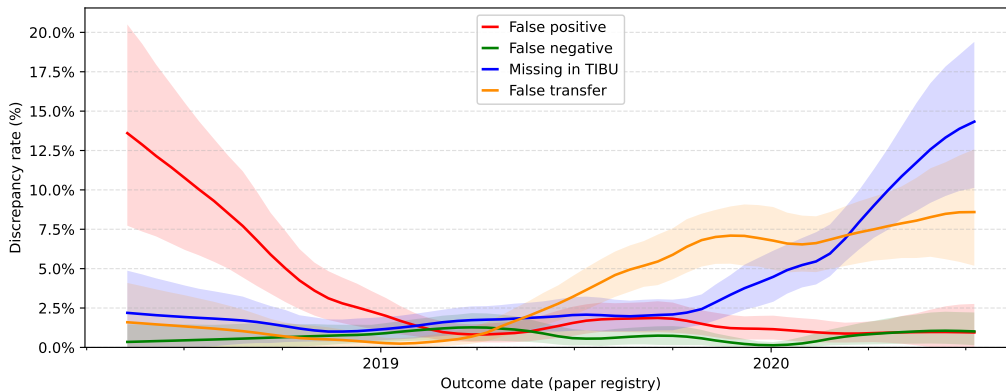


Figure 5: DQA discrepancy rates over time using paper-registry outcome dates throughout. Records without a parseable paper date are excluded. Compare with Figure 2.

C Discrepancy rates under the date-gated definition

This appendix recreates the breakdowns of Tables 3 and 4 and Figure 2 under the date-gated classification introduced in §2.3. A matched record is counted as a discrepancy only if the TIBU and paper outcomes disagree *and* the paper outcome date is strictly later than the TIBU outcome date; otherwise, on the reading that TIBU may reflect a post-audit paper update, the record is treated as concordant. Missing-in-TIBU is unchanged because TIBU is blank by construction. Records with an

unparseable date on either side preserve their headline classification ($n = 77$ false-positive and $n = 9$ false-negative records). Headline-vs-gated marginal totals are in Table 6.

Table 8: DQA discrepancy rates by province, urbanicity, and clinic size, under the date-gated definition. Compare with Table 3.

	<i>N</i>	(%)	False positive <i>N</i>	(%)	False negative <i>N</i>	(%)	Missing in TIBU <i>N</i>	(%)	False transfer <i>N</i>	(%)
<i>By province</i>										
Central	27	(0.8%)	0	(0.0%)	0	(0.0%)	13	(48.1%)	0	(0.0%)
Coast	708	(21.4%)	9	(1.3%)	3	(0.4%)	13	(1.8%)	6	(0.8%)
Eastern North	2	(0.1%)	0	(0.0%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Eastern South	264	(8.0%)	6	(2.3%)	2	(0.8%)	24	(9.1%)	12	(4.5%)
Nairobi	535	(16.2%)	11	(2.1%)	7	(1.3%)	4	(0.7%)	89	(16.6%)
North Eastern	821	(24.8%)	52	(6.3%)	4	(0.5%)	2	(0.2%)	14	(1.7%)
Nyanza North	392	(11.9%)	3	(0.8%)	5	(1.3%)	3	(0.8%)	1	(0.3%)
Nyanza South	40	(1.2%)	1	(2.5%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Rift Valley North	44	(1.3%)	0	(0.0%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Rift Valley South	425	(12.9%)	7	(1.6%)	3	(0.7%)	37	(8.7%)	4	(0.9%)
Western	31	(0.9%)	2	(6.5%)	0	(0.0%)	1	(3.2%)	0	(0.0%)
<i>By urban/rural</i>										
Urban	1,118	(33.8%)	16	(1.4%)	9	(0.8%)	21	(1.9%)	95	(8.5%)
Rural	2,167	(65.5%)	75	(3.5%)	15	(0.7%)	76	(3.5%)	31	(1.4%)
<i>By clinic size</i>										
Large clinic	1,273	(38.5%)	26	(2.0%)	8	(0.6%)	24	(1.9%)	89	(7.0%)
Small clinic	2,033	(61.5%)	65	(3.2%)	16	(0.8%)	90	(4.4%)	37	(1.8%)
All	3,306	(100.0%)	91	(2.8%)	24	(0.7%)	114	(3.4%)	126	(3.8%)

Note: Urban/rural classification is based on clinic records. Large clinics are those with at least 7 patients registered in TIBU (median); small clinics are those with fewer.

Table 9: DQA discrepancy rates by individual and disease characteristics, under the date-gated definition. Compare with Table 4.

	<i>N</i>	(%)	False positive <i>N</i>	(%)	False negative <i>N</i>	(%)	Missing in TIBU <i>N</i>	(%)	False transfer <i>N</i>	(%)
<i>Individual characteristics</i>										
Male	2,115	(64.0%)	53	(2.5%)	16	(0.8%)	81	(3.8%)	99	(4.7%)
<i>Age group</i>										
<15	458	(13.9%)	9	(2.0%)	3	(0.7%)	7	(1.5%)	11	(2.4%)
15–34	1,528	(46.2%)	47	(3.1%)	11	(0.7%)	43	(2.8%)	66	(4.3%)
35–54	965	(29.2%)	20	(2.1%)	8	(0.8%)	32	(3.3%)	42	(4.4%)
55+	331	(10.0%)	15	(4.5%)	2	(0.6%)	9	(2.7%)	7	(2.1%)
<i>Disease characteristics</i>										
Bacteriologically confirmed	2,326	(70.4%)	71	(3.1%)	15	(0.6%)	71	(3.1%)	102	(4.4%)
Extrapulmonary	367	(11.1%)	7	(1.9%)	2	(0.5%)	9	(2.5%)	16	(4.4%)
Retreatment	298	(9.0%)	12	(4.0%)	2	(0.7%)	5	(1.7%)	5	(1.7%)
HIV positive	729	(22.1%)	22	(3.0%)	11	(1.5%)	21	(2.9%)	21	(2.9%)
All	3,306	(100.0%)	91	(2.8%)	24	(0.7%)	114	(3.4%)	126	(3.8%)

Note: Each row shows the count and rate of records matching the row label, and the count and rate of each discrepancy type within those records. Rows are not mutually exclusive (e.g. a male HIV-positive patient appears in both rows).

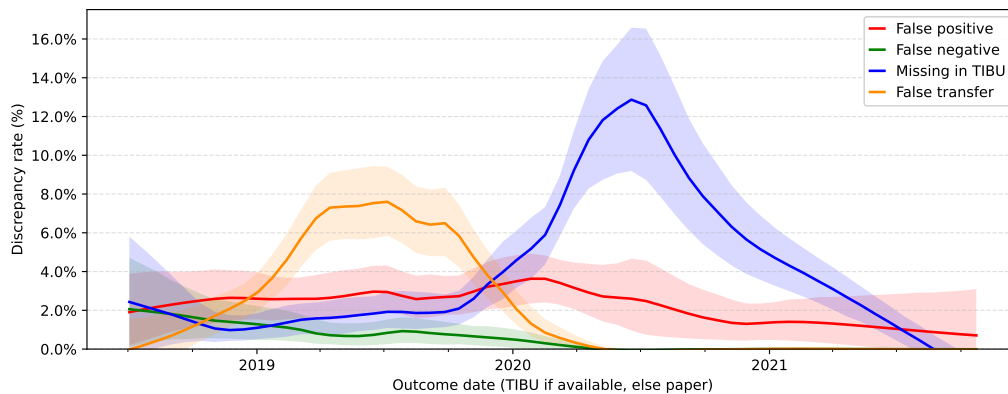


Figure 6: DQA discrepancy rates over time, under the date-gated definition. Compare with Figure 2.

D Multivariate predictors of missing-in-TIBU and false transfer

The marginal breakdowns in §§2.2.3 and 2.2.4 stratify discrepancy rates one characteristic at a time, leaving open whether apparent associations reflect that characteristic or a correlated confound. Table 10 reports logistic regressions of the two discrepancy categories with enough events to identify a multivariate model—missing-in-TIBU and false transfer—on individual, clinic, and record-age covariates jointly. (We model these two rather than false positives or false negatives because false negatives are rare (30 events) and false positives—though numerous (134)—are dominated by audit-snapshot artifacts that the date-gated analysis strips out (§2.3); their marginal evidence is in Tables 3–4.) Missing-in-TIBU is modelled with province fixed effects; the sparser false-transfer outcome is not, as too few false transfers occur per province to identify province effects (the cluster-robust covariance degenerates). For each outcome, the first column uses the full classifiable sample

and the second excludes that outcome's own two dominant clinics (Table 3b)—clinics 0 and 115 for missing-in-TIBU, clinics 162 and 50 for false transfer—as a sensitivity check, since events concentrated in a couple of clinics could otherwise drive coefficients on covariates merely correlated with their idiosyncratic profile.

Table 10: Average marginal effects on the probability of missing-in-TIBU and of false transfer, in percentage points; cluster-robust standard errors (by clinic) in parentheses.

	Missing in TIBU		False transfer	
	All	Excl. 0, 115	All	Excl. 162, 50
<i>Individual characteristics</i>				
Male	+2.79 (1.91)	-0.21 (0.51)	+2.17** (0.72)	+1.85 (1.02)
Age (years)	-0.00 (0.02)	-0.00 (0.01)	-0.04 (0.02)	-0.00 (0.01)
HIV positive	+0.17 (0.62)	+0.03 (0.31)	-1.15 (1.39)	-1.91*** (0.56)
Extrapulmonary TB	+1.01 (0.92)	+0.82 (0.61)	+0.82 (0.67)	-0.54 (0.75)
Retreatment case	+0.09 (0.81)	+0.07 (0.87)	-1.46 (2.07)	+0.17 (0.83)
Bacteriologically confirmed	+0.63 (0.81)	+0.98* (0.43)	+0.63 (1.30)	+0.06 (0.53)
<i>Clinic characteristics</i>				
Urban clinic	+1.61 (1.41)	+1.80 (0.96)	+0.71 (3.50)	+0.98 (1.31)
Clinic size (log patients)	-1.04 (0.58)	+0.19 (0.36)	+1.43** (0.51)	-1.09 (0.63)
<i>Record characteristics</i>				
Record age (years)	-8.43*** (2.35)	-3.44* (1.55)	-5.38** (2.04)	-3.21 (1.74)
Province fixed effects	Yes	Yes	No	No
<i>N</i>	3,180	2,776	3,180	2,771
Events	86	28	130	44
Clinics	53	51	53	51
McFadden's R^2	0.399	0.240	0.155	0.140

Notes: Cells report the average marginal effect on the probability of the column outcome, in percentage points; cluster-robust standard errors (by clinic), propagated to the AME via the delta method, are in parentheses below each estimate. Province fixed effects (included but not displayed) are identified for missing-in-TIBU but omitted for the sparser false-transfer outcome. Each “Excl.” column drops that outcome's two dominant clinics. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

For **missing-in-TIBU**, removing the two COVID clinics collapses both nominally interesting non-temporal coefficients. The point estimate on *Male* moves from +2.8 pp to −0.2 pp, and the estimate on *Clinic size* from −1.0 pp/log-patient to +0.2 pp; neither is statistically significant in either specification. *Record age* survives in both columns but its magnitude more than halves (from −8.4

to -3.4 pp/year, with significance falling from $p < 0.001$ to $p < 0.05$). We read this as a mostly mechanical reporting-lag effect: at any fixed snapshot date, more recent records have had less time to be transcribed into TIBU. The two COVID clinics inflate the slope because their missing-in-TIBU records cluster in the late, COVID-affected portion of the registration window. Outside those two clinics, neither patient demographics nor clinic-level structural features predict TIBU missingness; the only systematic signal is a modest record-age gradient consistent with reporting lag.

False transfer tells the same story as missingness, one clinic-set over. In the full sample three covariates reach significance—*Male* ($+2.2$ pp), *Clinic size* ($+1.4$ pp per log-patient), and *Record age* (-5.4 pp/year)—but all three collapse to non-significance once the two Nairobi clinics that drive false transfers (162 and 50) are excluded: Male falls to $+1.9$ pp, Clinic size flips to -1.1 pp, and Record age to -3.2 pp, none significant, while the event count falls from 130 to 44. (At 44 events that column is underpowered and should be read only for the disappearance of the full-sample signals, not for new ones—the lone significant coefficient there, on HIV status, is a small-sample artifact.) The apparent male and high-volume-clinic tendencies are therefore not a general labelling practice but the profile of those two clinics—just as the demographic gradients in missingness were the profile of clinics 0 and 115. Both discrepancy types are thus localized two-clinic phenomena rather than TIBU-wide patterns, each with its own pair of clinics.